

1

We create our bipartite graph with sets of vertices A, B each of size n

We say there is an edge from node a_i to b_j iff $i \neq j$

If the algorithm runs in the order of $a_1, b_1, a_2, \dots, b_{n-1}, a_n, b_n$ then we would end with n colors instead of 2

This is because for a_1 no colored neighbours so color is 1 and same with b_1 . For a_2 now, it has a colored neighbour that is b_1 which has a color of 1 so the next available color is 2. For a_i where $i < n$ all colors numbered less than i would be present in the neighbours of a_i so a_i would be colored i (same logic for b_i).

2

For showing that every k -chromatic graph has a k -critical induced subgraph. We can simply look at every induced subgraph of G that is still k -chromatic and pick the smallest one (or one that has no induced subgraphs in this set) and let's call it H . We know that $\chi(H) = k$ because it is in our defined set, and we know that removing any vertex from H will result in a graph smaller than H thus cannot be in our set since H is the smallest one.

Thus since our set was defined as k -chromatic graphs and induced subgraphs of our original graph G , we know that all induced subgraphs of H are induced subgraphs of G thus removing any vertex in H must result in a graph that is not k -chromatic. Therefore H is k -critical

As for proving the k -critical graph has minimum degree at least $k - 1$, originally I thought using Brook's theorem we get this part using the fact that $\Delta(G) \geq \chi(G)$ and the wrong part of my thinking was the if all vertices are of degree $k - 1$ then we have a complete graph which is not true!

Anyways, instead we do thing similar like in class with total colorings on planar graphs. Since H is k -critical we can remove a vertex and H should be only $k - 1$ colors. So proof by contradiction, suppose we have a vertex whose degree is less than $k - 1$, so a vertex v of degree $k - 2$.

Now if we remove v from H and color $H - v$ with our $k - 1$ colors, we can add back in v and since its degree is $k - 2$ we know there is at least one color available from our $k - 1$ colors backing $H - v + v$ $k - 1$ colorable which creates a contradiction because we started with $\chi(H) = k$

3

The two graphs of Brook's theorem that force high chromatic numbers are complete graphs and cycle graphs of odd length. Since we are avoiding cliques of size $\Delta(G) - 1$ we can look at altering the odd length cycle graphs

So odd cycle graphs have $\Delta = 2$ so by Brook's theorem $\chi = 3$. Since adding edges can only increase χ , if we add two edges to a vertex in the odd cycle graph to make our G such that $\Delta(G)$ is now 4 then we know that $\Delta(G) \geq \chi(C) = 3$ where C is the odd cycle graph.

However we need to add edges in a way that does not create a $\Delta(G) - 1 = 3$ clique. To ensure that we just make sure that our 2 new edges $[(i, j), (i, k)]$ have a distance greater than 2 between i, j, k without the edge (i, j) or (i, k) . To ensure this happens, suppose we use the odd cycle graph with 99 vertices and we add an edge from vertex 1 to vertex 33 and an edge from vertex 1 to vertex 66. This does not create a 3 clique and from previous points this graph has $\Delta(G) = 4$ and we know that $\chi(C_{99}) = 3$ so this graph meets the constraints specified in the problem.